

# Lung Nodule Segmentation — DynUNet / 3D U-Net — sliding-window ( $128^3$ ), CE weight 10

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## 1. Task

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Voxel-level segmentation of pulmonary nodules in 3D chest CT volumes, lung-bbox-cropped. Same task as the paper's `dynunet_v2` model — the differences are a reduced nodule CE weight (10 instead of 100) and the inference recipe: **sliding-window over  $128^3$  patches at native resolution** rather than a single forward on a  $256^3$  resample.

- **Input** (1, H, W, D) CT crop, intensity-normalised to [0, 1], lung-bbox + 20-voxel padding, kept at native voxel grid.
- **Training input**  $128^3$  patches sampled via `RandCropByPosNegLabeld` at a ratio of `pos:neg = 2:1`, `num_samples = 4` per volume.
- **Inference** MONAI `sliding_window_inference`, ROI (128, 128, 128), `overlap = 0.5`, `sw_batch_size = 4`, Gaussian blending, single output at the input resolution.
- **Output** (2, H, W, D) softmax logits: 0 = background, 1 = nodule.

## 2. Model

| Property                | Value                         |
|-------------------------|-------------------------------|
| Architecture family     | DynUNet / 3D U-Net            |
| Spatial dims            | 3                             |
| Input channels          | 1                             |
| Output channels         | 2 (softmax)                   |
| 6-level auto-configured | strides=[1,2,2,2,2], kernel=3 |
| Trainable parameters    | <b>16,538,530</b>             |

## 3. Data

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Same unified corpus + split as the paper's  $256^3$  baselines:

| Source          | Role               |
|-----------------|--------------------|
| NLST            | train + val + test |
| NSCLC-Radiomics | train + val + test |

| Source    | Role               |
|-----------|--------------------|
| LIDC-IDRI | train + val + test |

- **Split:** `unified_v2.json` — patient-grouped, dataset-stratified. 1 683 train / 297 val / 325 test series.
- **Lung crop:** per-series 3D bbox from `bboxes_unified.json` (lung ROI + medium-model fallback), + 20-vox padding. No resampling.
- **Class imbalance:** nodule voxels are  $\sim 10^{-5}$  of the total — see §7.

## 4. Training

| Setting         | Value                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Loss            | Focal Tversky + weighted CE ( $\alpha=0.3$ , $\beta=0.7$ , $\gamma=2.0$ , $\lambda_{\text{CE}}=0.1 / 0.3$ , nodule class weight = 10) |
| Optimizer       | Adam                                                                                                                                  |
| Learning rate   | $1 \times 10^{-5}$                                                                                                                    |
| Weight decay    | $1 \times 10^{-5}$                                                                                                                    |
| LR schedule     | Cosine annealing, $T_{\text{max}} = 400$ , $\eta_{\text{min}} = 1 \times 10^{-6}$                                                     |
| Batch size      | 2 volumes $\times$ 4 patches = 8 patches / step                                                                                       |
| Patch size      | $128^3$ (positive-biased random crop, <code>pos:neg = 2:1</code> )                                                                    |
| Epochs          | 400                                                                                                                                   |
| Random seed     | 42                                                                                                                                    |
| Mixed precision | bf16 (autocast, no GradScaler)                                                                                                        |
| Augmentation    | 3D flips, 90° rotations, elastic rotation, zoom, intensity scale/shift, Gaussian noise/blur, contrast                                 |
| Hardware        | 1 $\times$ NVIDIA H100 80 GB (fat01)                                                                                                  |

Best checkpoint recorded at **epoch 304 / 400** (see `_meta` in metrics JSON).

## 5. Evaluation protocol

Evaluated on the `unified_v2` **test** split (325 series held out — never seen during training or model selection). Predictions taken as `argmax` over the 2-channel softmax output from `sliding_window_inference`. All voxel-level metrics are micro-averaged over the whole split.

## 6. Results

### Ticket-required metrics

| Metric    | Value  |
|-----------|--------|
| mean IoU  | 0.6645 |
| Accuracy  | 0.9986 |
| Precision | 0.3634 |
| Recall    | 0.7839 |

### Supplementary

| Metric                          | Value  |
|---------------------------------|--------|
| IoU (foreground / nodule class) | 0.3303 |
| IoU (background class)          | 0.9986 |
| Dice / F1 (micro)               | 0.4966 |

### Per-case Dice distribution (325 test cases)

| Statistic | Value  |
|-----------|--------|
| Mean      | 0.4158 |
| Std       | 0.2599 |
| Min       | 0.0000 |
| p05       | 0.0058 |
| p25       | 0.1876 |
| Median    | 0.4209 |
| p75       | 0.6223 |
| p95       | 0.8231 |
| Max       | 0.9455 |

Voxels evaluated: 7,849,317,259. Cases: 325.

## 7. Notes on interpretation

Sliding-window inference at  $128^3$  trades global context (a  $256^3$  resample sees the whole lung in one pass) for preserved resolution (no down-sampling of small nodules). On this unified corpus the trade tends to be neutral-to-negative on nodule Dice — small-nodule gains do not compensate for lost global context on typical test cases.

Class-imbalance caveats are unchanged from the 256<sup>3</sup> reports: Accuracy is trivially ~1, mIoU is dominated by IoU\_bg, and Dice / F1 (micro) is the honest voxel-level summary. Per-case mean weights each patient equally regardless of nodule volume.

## Instance-level metrics (per-nodule)

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Connected-component analysis with 26-connectivity, pooled over the held-out test split, on the native voxel grid.

- **Instance recall** — GT nodules touched by at least one predicted voxel, over all GT nodules.
- **Instance precision** — predicted components touching at least one GT voxel, over all predicted components.

| Metric                          | Value         |
|---------------------------------|---------------|
| GT nodules (components)         | 947           |
| GT nodules hit                  | 820           |
| <b>Instance recall</b>          | <b>0.8659</b> |
| Predicted components            | 7,686         |
| Predicted components hitting GT | 839           |
| <b>Instance precision</b>       | <b>0.1092</b> |

No minimum-size filtering is applied — every predicted component counts, so single-voxel false positives lower instance precision. Produced by `scripts/instance_metrics.py --config configs/dynunet_sw_ce10.yaml --ckpt <ckpt> --split test` (JSON: `results_instance/dynunet_sw_ce10.json`).

## 8. Reproducibility

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- Config `/home/werner/nodule-segmentation/configs/dynunet_sw_ce10.yaml`
- Checkpoint `stage2_fine/checkpoints_dynunet_sw_ce10/best_model.pth`
- Metrics JSON `/tmp/claude-1002/-home-werner-nodule-segmentation/c2faa0a0-5a75-48f5-8288-377df40092b7/scratchpad/ce10_test/dynunet.json`
- Command `python stage2_fine/eval_metrics.py --config /home/werner/nodule-segmentation/configs/dynunet_sw_ce10.yaml --checkpoint stage2_fine/checkpoints_dynunet_sw_ce10/best_model.pth --split test`